

Information Extraction: NER (cont) + Relation Extraction

Carolina Scarton

`c.scarton@sheffield.ac.uk`

Text Processing
Department of Computer Science, University of Sheffield
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Learning objectives

- ▶ Summarise and apply supervised learning for NER
- ▶ Define the Entity Linking approach
- ▶ Define the Relation Extraction task
- ▶ Explain knowledge engineering and supervised learning approaches for relation extraction

[Information Extraction]

Named Entity Recognition

Information Extraction: NER – recap

Entity Extraction/Named Entity Recognition (NER)

Task: Identify the **extent** and the **type** of each textual mention of an entity

The set of types is determined in advance (e.g. organisation, person, date, etc...)

Cable and Wireless	today announced ...	Extent: 0-3 ; Type = ORG
IBM	and Microsoft	today announced ...
		Extent: 0-1 ; Type = ORG
		Extent: 2-3 ; Type = ORG
John Lewis	hired ...	Extent: 0-2 ; Type = ORG
Theresa May	hired.	Extent: 0-2 ; Type = PER

Information Extraction: NER – recap

Types of entities addressed by IE systems include:

► Named individuals

- Organisations (ORG), persons (PER), books, films, ships, restaurants ...

→ Cable and Wireless today announced ...; Extent: **0-3** ; Type = **ORG**

→ Barack Obama was the 44th president... ...; Extent: **0-2** ; Type = **PER**

- Geo-Political entities (GPE), locations (LOC)

→ The Mont Blanc intersects France, Italy and Switzerland. ; Extent: **1-3** ; Type = **LOC**

→ The Mont Blanc intersects France , Italy and Switzerland . ; Extent: **4-5** ; Type = **GPE**

► Times: temporal expressions dates, times of day

→ Let's meet at 2pm next Friday ...; Extent: **3-4** ; Type = **TIME**

→ Let's meet at 2pm next Friday ...; Extent: **5-6** ; Type = **DATE**

► Measures: monetary expressions, distances/sizes, weights ...

→ This watch costs £35 ...; Extent: **3-4** ; Type = **MONEY**

NER: Knowledge Engineering Approaches – recap

System has four main stages:

1. **Lexical processing**

- pre-processing (tokenisation, etc)
- gazetteer lookup
- trigger word tagging

2. **NE parsing**

- syntactic-motivated rules (e.g. ORGAN_NP → NAMES_NP & NAMES_NP)

3. Discourse interpretation - **Coreference Resolution**

4. Discourse interpretation - **Semantic Type Resolution**

NER: Supervised Learning

Example:

[American Airlines]_{ORG}, a unit of [AMR Corp.]_{ORG}, immediately matched the move, spokesman [Tim Wagner]_{PER} said.

[Source: Jurafsky and Martin, 2nd ed., p. 730]

NER: Supervised Learning

Example:

[American Airlines]_{ORG}, a unit of [AMR Corp.]_{ORG}, immediately matched the move, spokesman [Tim Wagner]_{PER} said.

[Source: Jurafsky and Martin, 2nd ed., p. 730]

In **BIO** encoding this example looks like this:

American	B _{ORG}
Airlines	I _{ORG}
a	O
unit	O
of	O
AMR	B _{ORG}
Corp.	I _{ORG}
,	O
immediately	O
matched	O
the	O
move	O
,	O
spokesman	O
Tim	B _{PER}
Wagner	I _{PER}
said	O
.	O

NER: Supervised Learning

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In **BIO** encoding this example looks like this:

Supervised ML technique:

- ▶ Trained on a corpus of labelled sequences, like this
- ▶ Aim: predict the labelling of a new, unlabelled example.



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Airlines	I _{ORG}
a	O
unit	O
of	O
AMR	B _{ORG}
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,	O
immediately	O
matched	O
the	O
move	O
,	O
spokesman	O
Tim	B _{PER}
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.	O

NER: Supervised Learning

Features for Sequence Labelling

Each token is typically represented as a set of features:

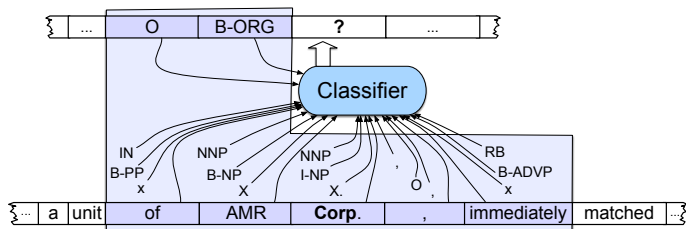
- ▶ the **token** itself
- ▶ characteristics of the **token**: stem, Part-Of-Speech, etc.
- ▶ characteristics of the **neighbouring tokens** (window of ± 2 or 3 tokens either side)

Typical features used: [Source: Jurafsky and Martin, 2nd ed., p. 731]

Features	Explanations	Ex.: "The shop L'Occitane en Provence"
Lexical items	The token to be labeled	L'Occitane
Stemmed lexical items	Stemmed version of the target token	Occitan
Shape	Orthographic pattern of the target token	X'Xxxxxxx or X'Xx
Character affixes	Character-level affixes	L, L', L'O, L'Oc, tane, ane, ne, e
Part of Speech	Part of speech of the token	NNP
Syntactic chunk labels	Base-phrase chunk label	B-NP
Gazetteer or name list	Token seen in a named entity lists	-
Predicted token(s)	Previously predicted classes	O O
Bag of words/Bag of n-grams	Tokens and/or n-grams in the context	The, shop, The shop, etc.

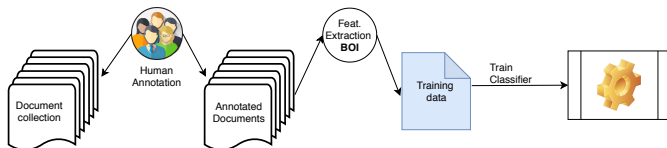
NER: Supervised Learning

Graphical representation of features used to NER:

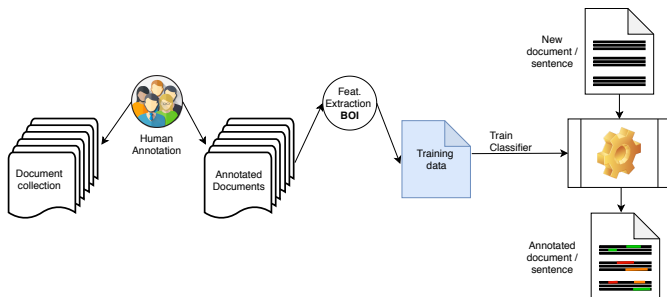


[Source: Jurafsky and Martin, 2nd ed., p. 733]

NER: Supervised Learning



NER: Supervised Learning



NER: Supervised Learning

BIO sequence labelling: Carreras et al. (2003)

[Carreras et al., 2003]

→ best score at CONLL 2003 NER shared task

NER: Supervised Learning

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→ best score at CONLL 2003 NER shared task

Use of a two-pass method:

1. **NE detection**: assign un-typed **BIO** tags → find NE boundaries **whatever the class**
2. **NE classification**: assign a class to all **BI** tags

NER: Supervised Learning

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Use of additional features:

- ▶ Type pattern of consecutive word in context: functional (f), capitalized (C), lowercased (l) etc. → "John Smith payed 3 euros" ⇒ CC|xl

NER: Supervised Learning

BIO sequence labelling: Carreras et al. (2003)

[Carreras et al., 2003]

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Use of **Adaboost** classifier:

- ▶ **Adaptive Boosting** is a meta-algorithm that combines outputs of other learning algorithms (called **weak learners**). Subsequent weak learners are tweaked in favor of the instances that are misclassified by previous classifiers.

NER: Supervised Learning

BIO sequence labelling: Carreras et al. (2003)

[Carreras et al., 2003]

	English		German	
	Precision (%)	Recall (%)	Precision (%)	Recall (%)
NE Detection	91.93	94.02	85.85	72.61
NE Classification	84.05	85.96	75.47	63.82
	Accuracy		Accuracy	
Oracle	95.14		85.14	

Oracle: Classification with a perfect Detection

⇒ LOC and PER score consistently higher than ORG and MISC

Information Extraction: NER recap → Supervised Learning

Example: apply the BIO annotation

Despite stars Kevin Hart and Dwayne 'The Rock' Johnson having very busy schedules, producer Hiram Garcia has hinted that things are already underway in Hollywood.

Information Extraction: NER recap → Supervised Learning

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Despite stars [Kevin Hart]_{PER} and [Dwayne 'The Rock' Johnson]_{PER} having very busy schedules, producer [Hiram Garcia]_{PER} has hinted that things are already underway in [Hollywood]_{LOC}.

Information Extraction: NER recap → Supervised Learning

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Despite stars [Kevin Hart]_{PER} and [Dwayne 'The Rock' Johnson]_{PER} having very busy schedules, producer [Hiram Garcia]_{PER} has hinted that things are already underway in [Hollywood]_{LOC}.

[Despite]_O [stars]_O [Kevin]_{B_PER} [Hart]_{I_PER} [and]_O [Dwayne]_{B_PER} [']_{I_PER} [The]_{I_PER} [Rock]_{I_PER} [']_{I_PER} [Johnson]_{I_PER} [having]_O [very]_O [busy]_O [schedules]_O [,]_O [producer]_O [Hiram]_{B_PER} [Garcia]_{I_PER} [has]_O [hinted]_O [that]_O [things]_O [are]_O [already]_O [underway]_O [in]_O [Hollywood]_{B_LOC}.

Information Extraction: NER recap → Supervised Learning

Example: apply the BIO annotation

William MacDonald took the bus from Kelham Island to Manor Park, aiming to visit the Sheffield Manor Lodge.

Information Extraction: NER recap → Supervised Learning

Example: apply the BIO annotation

William MacDonald took the bus from Kelham Island to Manor Park, aiming to visit the Sheffield Manor Lodge.

[William MacDonald]_{PER} took the bus from [Kelham Island]_{LOC} to [Manor Park]_{LOC}, aiming to visit the [Sheffield Manor Lodge]_{LOC}.

Information Extraction: NER recap → Supervised Learning

Example: apply the BIO annotation

William MacDonald took the bus from Kelham Island to Manor Park, aiming to visit the Sheffield Manor Lodge.

[William MacDonald]_{PER} took the bus from [Kelham Island]_{LOC} to [Manor Park]_{LOC}, aiming to visit the [Sheffield Manor Lodge]_{LOC}.

[William]_{B_PER} [MacDonald]_{I_PER} [took]_O [the]_O [bus]_O [from]_O [Kelham]_{B_LOC} [Island]_{I_LOC} [to]_O [Manor]_{B_PER} [Park]_{I_LOC} [,]_O [aiming]_O [to]_O [visit]_O [the]_O [Sheffield]_{B_LOC} [Manor]_{I_LOC} [Lodge]_{I_LOC}.

Information Extraction: Overview

- ▶ **Introduction to Information Extraction**
 - ▶ Definition + contrast with IR
 - ▶ Example Applications
 - ▶ Overview of Tasks
 - ▶ Overview of Approaches
 - ▶ Evaluation + Shared Task Challenges
 - ▶ Brief(est) history of IE
- ▶ **Named Entity Recognition**
 - ▶ Task
 - ▶ Approaches: Rule-based, Supervised Learning
 - ▶ **Entity Linking**
- ▶ **Relation Extraction**
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 - ▶ Approaches: Rule-based, Supervised Learning, Bootstrapping, Distant Supervision

NER: Entity Linking

IE using knowledge base population (**KBP**)

- ▶ facts are gathered from open access web sources
- ▶ then used to build a structured information repository

⇒ To correctly assemble facts, the entities must be linked to the appropriately entry in the KB

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Entity Linking Task:

Given a text with a **recognised NE mention** in that text and a **knowledge base (KB)**, such as Wikipedia, link the NEs to the matching entry in the KB if there is one, else create an entry.

NER: Entity Linking

Is this task difficult? – yes!!

- ▶ Wikipedia contains over 200 entries for **John Smith**, at least 1,716 places called **San José** (or San Jose)

[Source: Wikipedia]

Explorer [[edit](#)]

- John Smith (explorer) (1580–1681), helped found the Virginia Colony and became Colonial Governor of Virginia

Academics [[edit](#)]

- John Smith (anatomist and chemist) (1721–1797), professor of anatomy and chemistry at the University of Oxford, 1768–97
- John Blair Smith (1764–1799), president of Union College, New York
- John Smith (Cambridge, 1768), vice chancellor of the University of Cambridge, 1768 until 1767
- John Smith (astronomer) (1774–1805), Lucasian Professor of Astronomy and Master of Caius
- John Smith (zooarchaeologist) (died 1920), professor of languages at Dartmouth College
- John Augustine Smith (1780–1866), president of the College of William and Mary, 1814–1826
- John Smith (botanist) (1780–1800), curator of Kew Gardens
- John Smith (physician) (c.1600–1676), Scottish physician specialising in treating the insane
- John Alexander Smith (physician) (1816–1883), Scottish physician, antiquarian, archaeologist and ornithologist
- J. Lawrence Smith (1818–1883), American doctor and chemist
- John Smith (dentist) (1825–1910), founder of Edinburgh's School of Dentistry
- John Campbell Smith (1828–1914), Scottish writer, advocate and Sheriff-Substitute of Forfarshire
- John Donnell Smith (1829–1908), biologist and taxonomist
- John McCarville Smith (1844–1918), Australian metallurgist and bacteriologist
- John Alexander Smith (1863–1909), British bioidal philosopher
- John Maynard Smith (1920–2004), geneticist
- John Cyril Smith (1922–2002), leading authority on English criminal law
- John Derek Smith (1924–2005), Cambridge molecular biologist
- John Smith (sociologist) (1927–2002), English sociologist
- John D. Smith (born 1946), biologist at the University of Cambridge
- John H. Smith (mathematician), Boston College educator (retired 2005)

Arts [[edit](#)]

- John Smith (engraver) (1552–1742), English mazzettini engraver
- John Smith (English poet) (1602–1717), English poet and playwright
- John Christopher Smith (1713–1746), English composer
- John Warwick Smith (1746–1801), British watercolour landscape painter and illustrator
- John Stafford Smith (1750–1836), composer of the tune for "The Star-Spangled Banner"
- John Raphael Smith (1751–1812), English miniature engraver and painter
- John Thomas Smith (engraver) (1756–1832), draughtsman, engraver and antiquarian
- John Smith (clockmaker) (1770–1816), Scottish clockmaker
- John Rubens Smith (1770–1849), London born painter, printmaker and art instructor who worked in the United States
- John Smith (architect) (1781–1862), Scottish architect
- John Smith (art historian) (1781–1855), British art dealer
- John Otin Smith (1790–1843), English wood engraver
- John Frederick Smith (1806–1860), English novelist
- John Rowson Smith (1810–1866), American romance painter
- John G. Smith (poet) (died 1881), Scottish poet
- John Myer Smith (1830–1912), British artist and designer
- John Berryman (1914–1972), originally John Allyn Smith, American poet
- John Smith (Canadian poet) (born 1927), Canadian poet
- John Smith (actor) (1924–1980), American actor
- John H. Smith (born 1942), Canadian film director and screenwriter
- John Templeton Smith (born 1943), British-born American thriller novelist
- John Smith (English filmmaker) (born 1952), avant-garde filmmaker
- John Warner Smith (born 1952), American poet and educator
- John F. Smith, American soap opera writer
- John Smith (comic strip writer) (born 1967), British comic writer
- John Smith (musician), English contemporary folk musician and recording artist

Religion [[edit](#)]

- John Smith (Bishop of Landaff) (died 1478), Welsh bishop
- John Smith (Psalterist) (1618–1652), one of the founders of the Cambridge Psalterists
- John Smith (Bishop of Kilkee and Achonry) (died 1686), Irish Anglican priest
- John Smith (Unitarian) (fl. 1648–1725), Unitarian writer
- John Smith (priest, born 1660) (1660–1715), English writer of Bells
- John Smith (uncle of Joseph Smith) (1781–1854), Presiding Patriarch and member of the First Presidency of The Church of Jesus Christ of Latter-day Saints
- John Smith (Restoration Movement) (1784–1808), early Restoration Movement leader
- John Smith (monastery) (1790–1856), English monastery in Denmark
- John Smith (Revolutionist) (1794–1831), English Methodist minister known as "The Revivalist"
- John Smith (Anglican priest) (1798–1870), first person to transcribe the Diary of Samuel Pepys
- John Smith (nephew of Joseph Smith) (1802–1811), Presiding Patriarch of The Church of Jesus Christ of Latter-day Saints
- John Henry Smith (1804–1811), apostle in The Church of Jesus Christ of Latter-day Saints
- John Taylor Smith (1800–1838), Anglican Bishop of Sierra Leone
- John Smith (Archdeacon of Wilton) (1820–2002), Anglican priest
- John H. Smith (priest) (1825–2016), American Roman Catholic bishop
- John H. Smith (bishop) (1864–2012), American Episcopal priest, Bishop of the Episcopal Diocese of West Virginia, 1989–1999
- John Smith (Buck's Squid) (died 2018), Australian founder of the Buck's Squid enterprise club

Sports [[edit](#)]

Cricketer [[edit](#)]

- John Smith (Barnfield cricketer), English cricketer
- John Smith (cricketer, born 1833) (1833–1906), Lancashire and Yorkshire cricketer
- John Smith (cricketer, born 1834) (1834–?), Nottinghamshire and All-England Eleven cricketer
- John Smith (cricketer, born 1835) (1835–1888), English cricketer
- John Smith (Derbyshire cricketer) (1841–1888), English cricketer, played for Derbyshire
- John Smith (cricketer, born 1843) (1843–1873), English cricketer, played for Cambridgeshire 1862–72
- John Smith (cricketer, born 1860) (1860–1946), English cricketer, played for Leicestershire 1921
- John Smith (cricketer, born 1906) (1906–1971), English cricketer, played for Leicestershire 1930–1959
- John Smith (cricketer, born 1936) (born 1936), Australian cricketer
- John Smith (New Zealand cricketer) (born 1965), New Zealand cricketer also known as Campbell Smith

Association football [[edit](#)]

- John Smith (footballer, born 1855) (1855–1934), Scottish footballer of the 1870s and 1880s
- John Smith (footballer, born 1862) (1862–1911), Scottish footballer who played as a winger
- John Smith (footballer, born 1866) (1866–?), Scottish footballer (for United, Manchester, Scotland)
- John Smith (footballer, born 1907), English footballer for Bradford City and Blackburn Rovers
- John Smith (footballer, born 1921) (1921–?), English footballer
- John Smith (footballer, born 1927) (1927–2005), English footballer (Leicester)
- John Smith (footballer, born 1928) (1928–1996), West Ham United
- John Smith (footballer, born 1944), Welsh footballer (Sheffrham)
- John Smith (footballer, born 1973), football full back (Thames Valley)
- John Smith (footballer, born 1977), retired English footballer

Criminals [[edit](#)]

- John Smith (housebreaker) (1661–after 1727), London housebreaker who evaded hanging three and was eventually transported to Virginia
- John Smith (murderer) (born 1951), American convicted murderer who killed his first and second wives
- John Eldon Smith (1830–1883), convicted of the murders of Ronald and Juana Nola

Businessmen [[edit](#)]

- John Smith (brewer) (1804–1876), Tadcaster brewery founder in North Yorkshire, UK
- John F. Smith, Jr. (born 1938), former chairman and chief executive officer, General Motors
- John L. Smith (1923–1990), African American abolitionist, Underground Railroad confidant and politician
- John L. Smith (pharmaceutical executive) (1889–1995), German-born American chemist, pharmaceutical executive and Major League Baseball team owner
- John Smith (executive) (born 1967), former chief executive officer, BBC Worldwide Ltd
- John Russell Smith (1810–1894), English bookbinder and bibliographer

Others [[edit](#)]

- John Smith (Chippewa Indian) (died 1602), reported to have died at the age of 137

NER: Entity Linking: Approaches

- ▶ Simple idea: given a text T containing an NE mention m and using Wikipedia as a KB
 1. index all pages in the KB using IR
 2. build a query from T containing m + search the KB
 3. pick the best ranked page from step 2
- does not work very well

NER: Entity Linking: Approaches

- ▶ Simple idea: given a text T containing an NE mention m and using Wikipedia as a KB
 1. index all pages in the KB using IR
 2. build a query from T containing m + search the KB
 3. pick the best ranked page from step 2
- does not work very well
- ▶ More successful approaches consider disambiguating all NEs jointly
 - ▶ **Intuition:** in disambiguating a text mentioning **Ashoka** and **Sheffield**, the **Ashoka** mentioned is likely to be in Sheffield, while the Sheffield is likely to be one containing an Ashoka restaurant.
 - ▶ See, e.g., Alhelbawy and Gaizauskas (2014)

NER: Conclusion

Named Entity Recognition (NER) is a core IE technology

- ▶ detect and classify all mentions of NE of a given set of entity types within a given text
- ▶ now relatively mature and at “usable” performance levels

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Techniques used have included:

- ▶ knowledge engineering approaches
- ▶ supervised learning approaches, e.g. BIO sequence labelling

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Open challenges include:

- ▶ reducing the amount of training data needed via, e.g. bootstrapping techniques
- ▶ exploiting existing structured data sources to generate “weakly labelled” training data (aka distant supervision)
- ▶ expanding the classes of entities addressed
- ▶ developing NERs for languages other than English

NER: Conclusion

GATE platform has multiple NER tools

- ▶ support for multiple domains (e.g. Twitter, Biomedical)
- ▶ multiple languages (e.g. English, Russian, German, Welsh)

`https://cloud.gate.ac.uk`

[Information Extraction] Relation Extraction

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Relation Extraction

- ▶ **Task definition**

Approaches to Relation Extraction

- ▶ Knowledge-engineering approaches to RE
- ▶ Supervised learning approaches to RE
- ▶ Bootstrapping Approaches to RE
- ▶ Distant Supervision Approaches to RE

Information Extraction: Relation Extraction – recap

Relation Extraction (RE)

Task: Identify all **assertions of relations** holding between **entities** in a text T

- ▶ the entities are identified in previous entity extraction step
- ▶ The set of possible relations **R** is determined in advance

Information Extraction: Relation Extraction – recap

Relation Extraction (RE)

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Note:

- ▶ relations in $\mathbf{R}$ are usually binary
- ▶ the entity classes involved in a relation in $\mathbf{R}$ are assumed to be a subset of those identified in the entity extraction process

Information Extraction: Relation Extraction – recap

Relation Extraction (RE)

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Generally divided into two subtasks:

1. **Relation detection**: find pairs of entities between which a relation holds
2. **Relation classification**: determine the type of a previously detected relation

Information Extraction: Relation Extraction: Examples

Some examples

- ▶ LOCATION_OF holding between:
 - ▶ ORGANISATION and GEOPOLITICAL_LOCATION
 - ▶ medical INVESTIGATION and BODY_PART
 - ▶ GENE and CHROMOSOME_LOCATION
- ▶ EMPLOYEE_OF holding between PERSON and ORGANISATION
- ▶ PRODUCT_OF holding between ARTIFACT and ORGANISATION
- ▶ IS_EXPOSED_TO holding between ORGANISATION and RISK
- ▶ IS_ASSOCIATED_WITH holding between DRUG and SIDE_EFFECT
- ▶ INTERACTION holding between PROTEIN and PROTEIN

Information Extraction: RE: Challenges

Challenges for Relation Extraction

1. Many **different ways** to express the same relation
 - ▶ **Canonical:** [Microsoft]_{ORG} is located in [Redmond]_{LOC}
 - ▶ **Synonyms:** [Microsoft]_{ORG}
is located/based/headquartered in [Redmond]_{LOC}
 - ▶ **Syntactic variations:**
 - ▶ [Microsoft]_{ORG}, the software giant and ... , is based in [Redmond]_{LOC}
 - ▶ [Redmond]_{LOC}-based [Microsoft]_{ORG} ...
 - ▶ [Redmond]_{LOC}'s [Microsoft]_{ORG} ...
 - ▶ [Redmond]_{LOC} software giant [Microsoft]_{ORG} ...

Information Extraction: RE: Challenges

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 - ▶ [Redmond]_{LOC}'s [Microsoft]_{ORG} ...
 - ▶ [Redmond]_{LOC} software giant [Microsoft]_{ORG} ...
2. Relations often involve **coreference links**

[Bill Gates]_{PER} is co-founder, technology advisor and board member of [Microsoft]_{ORG}. [He]_{PER} served as chairman of the board until Feb. 4, 2014. On June 27, 2008, [Gates]_{PER} transitioned out of a day-to-day role in the company to spend more time on his global health and education work at the [Bill & Melinda Gates Foundation]_{ORG}.

Information Extraction: RE: Challenges

Challenges for Relation Extraction

3. The information required may be **spread across multiple sentences**

Information Extraction: RE: Challenges

Challenges for Relation Extraction

3. The information required may be **spread across multiple sentences**
4. The information may be implied by the text rather than explicitly asserted → **inference**

Ex1.: **[Dirk Ruthless]**_{PER} of [MegaCorp]_{ORG} made a stunning announcement today. In September **[he]**_{PER} will be stepping down as [Chief Executive Officer]_{POS} to spend more time with his pet piranhas.

- ⇒ resolve **pronominal anaphor** **Dirk Ruthless** ⇔ **he** to determine the corporate POSITION
- ⇒ In Ex1.: no explicit statement that **Dirk Ruthless** is CEO of MegaCorp
 - ▶ "[**Dirk Ruthless**]_{PER} of [MegaCorp]_{ORG}" + "will be stepping down as [Chief Executive Officer]_{POS}" = **[Dirk Ruthless]**_{PER} is CEO of [MegaCorp]_{ORG}

Information Extraction: RE: Challenges

Challenges for Relation Extraction

3. The information required may be **spread across multiple sentences**
4. The information may be implied by the text rather than explicitly asserted → **inference**

Ex1.: **[Dirk Ruthless]**_{PER} of [MegaCorp]_{ORG} made a stunning announcement today. In September **[he]**_{PER} will be stepping down as [Chief Executive Officer]_{POS} to spend more time with his pet piranhas.

- ⇒ resolve **pronominal anaphor** **Dirk Ruthless** ⇔ **he** to determine the corporate POSITION
- ⇒ In Ex1.: no explicit statement that **Dirk Ruthless** is CEO of MegaCorp
 - ▶ "[**Dirk Ruthless**]_{PER} of [MegaCorp]_{ORG}" + "will be stepping down as [Chief Executive Officer]_{POS}" = **[Dirk Ruthless]**_{PER} is CEO of [MegaCorp]_{ORG}
 - ▶ Solving RE may imply solving **textual entailment**

Information Extraction: Relation Extraction: Overview

Relation Extraction

- ▶ Task definition

Approaches to Relation Extraction

- ▶ **Knowledge-engineering approaches to RE**
- ▶ Supervised learning approaches to RE
- ▶ Bootstrapping Approaches to RE
- ▶ Distant Supervision Approaches to RE

RE: Knowledge Engineering

[Mr. Wright]_{PER}, [executive vice president]_{POSITION} of [Merrill Lynch Canada Inc.]_{ORG}

Such systems use manually authored rules and can be divided into

- ▶ “**shallow**”: systems engineered to the IE task, typically using **pattern-action** rules

Pattern: **“\$Person, \$Position of \$Organization”**

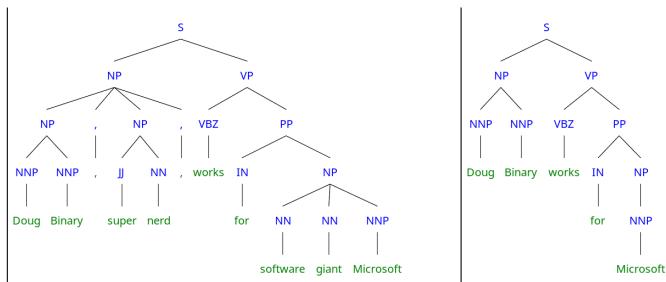
Action: **add-relation(is-employed-by(\$Person,\$Organization))**

RE: Knowledge Engineering

- ▶ “**deep**”: linguistically inspired language understanding systems
 - ▶ **parse input** to identify key grammatical relations (e.g. subject, object)
 - ▶ use **transduction rules** on parser output to extract relations
 - **more powerful than regex on NE tags alone**

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- ▶ Ex. parse trees:
 - ▶ Multiple surface forms share underlying syntactic structure
 - ▶ SUBJECT = PER, OBJECT = ORG and VERB = **works for**

Strengths

- ▶ High precision
- ▶ Interpretable results → system behaviour is human-comprehensible

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Weaknesses

- ▶ The writing of rules has no end
- ▶ New rules needed for every new domain
 - ▶ pattern action rules for shallow approaches
 - ▶ transduction rules for deep approaches

Information Extraction: Relation Extraction: Overview

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Information Extraction: RE: Supervised learning

Question: what is to be learned?

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- ▶ Classifies a sentence as to whether it bears a **specific relation between some entity types**
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 - ▶ **relation classification**: determines the relation (multi-way)

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- ▶ Can be divided in two stages:
 - ▶ **relation detection**: determines whether a sentence expresses a relation (binary)
 - ▶ **relation classification**: determines the relation (multi-way)
- ▶ Rule learning popular until early 2000's. Then classifier approach.

→ Details on classifier approach only

Information Extraction: RE: Supervised Learning

Classification approaches to relation extraction:

- ▶ Assume entities are already tagged → **output of NER**
- ▶ Use an algorithm to learn **binary classifiers to distinguish instances** where
 - ▶ entities co-occur and relation holds (**positive instances**)
 - ▶ entities co-occur and relation does not hold (**negative instances**)

Information Extraction: RE: Supervised Learning

Key issue: what **features** to use to represent instances?

- ▶ 3 broad classes:
 - ▶ Features of the **Named Entities**
 - ▶ Features from the **words in the text**
 - ▶ words between the two NE
 - ▶ words surrounding the candidates (left of 1st word and right of 2nd word)
 - ▶ Features about the **entity pair within the sentence**, e.g.
 - ▶ distance between the entities (in words or constituents)
 - ▶ is there a NE in between them?
 - ▶ clues from the syntactic structure of the sentence (parse tree)

Information Extraction: RE: Supervised Learning: Example

[American Airlines]_{ORG}, a unit of [AMR Inc.]_{ORG}, immediately matched the move, spokesman [Tim Wagner]_{PERS} said.

[Source: Jurafsky and Martin, 2nd ed., p. 730]

Features extracted from this example:

Features	Value
Entity-based features	
Entity ₁ type	ORG
Entity ₁ head	<i>airlines</i>
Entity ₂ type	PERS
Entity ₂ head	<i>Wagner</i>
Concatenated types	ORGPERS
Word-based features	
Between-entity BOW	{ <i>a, unit, of, AMR, Inc., immediately, matched, the, move, spokesman</i> }
Word(s) before Entity ₁	NONE
Word(s) before Entity ₂	<i>said</i>
Syntactic features	
Constituent path	NP ↑ NP ↑ S ↑ S ↓ NP
Base syntactic chunk path	NP → NP → PP → NP → VP → NP → NP
Typed-dependency path	<i>Airlines</i> ← _{subj} <i>matched</i> ← _{comp} <i>said</i> → _{subj} <i>Wagner</i>

[Source: Jurafsky and Martin, 2nd ed., p. 730]

Information Extraction: RE: Supervised Learning

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- ▶ No need to write extensive/complex rule sets for each domain
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- beware of data sparsity!

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Weaknesses

- ▶ Quality of relation extraction dependent on quality and quantity of training data
- can be difficult, costly and time consuming to generate
- ▶ Feature extractors can be noisy (e.g. parsers) → reduce overall performance

Reading material

- ▶ Daniel Jurafsky & James H. Martin (2021).
Chapter 17: Information Extraction.
Speech and Language Processing, 3rd ed. draft.
<https://web.stanford.edu/~jurafsky/slp3/17.pdf>

Credit: slides based on Dr Loïc Barrault's slides for the autumn 2020/2021 course.

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